Synopsis

AF 52

on

Custom LLM for IOFT's Humanoid Robot

Submitted in partial fulfilment of the requirements of the degree of

Bachelor of Technology

In

Electronics and Computer Science Engineering

by

Smit Patel [EU2237021]

Swikar Patil [EU2237006]

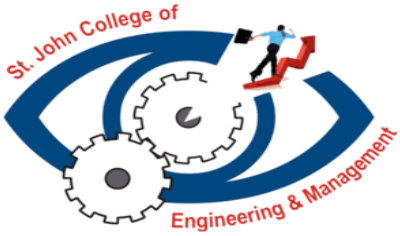
Aaditya Singh [EU1237012]

Ayush Varma [EU2237015]

Under the Guidance of

Mrs. Vishakha Rane

Assistant Professor, EXCS, SJCEM



DEPARTMENT OF ELECTRONICS AND COMPUTER SCIENCE

ST. JOHN COLLEGE OF ENGINEERING & MANAGEMENT

UNIVERSITY OF MUMBAI

2026-2027

St. John College of Engineering and Management
Autonomous Institute
(A Christian Religious Minority Institution)
Approved by AICTE and DTE, Affiliated to University of Mumbai / MSBTE
DTE Code : 3218 AICTE Permanent ID : 1-4790201
NAAC Accredited with Grade 'A+', Three Programs NBA Accredited

Certificate of Approval

This is to certify that **Smit Patel, Swikar Patil, Aaditya Singh, and Ayush Varma are bona fide students of the B. Tech** program in **Electronics and Computer Science** Department of **St. John College of Engineering and Management**, **Mumbai**. They have satisfactorily completed the requirements of **Synopsis** as prescribed by **ST. JOHN COLLEGE OF ENGINEERING AND MANAGEMENT (An Autonomous College affiliated to University of Mumbai),** while working on "**Custom LLM for IOFT's Humanoid Robot".**

Guide Name: **Mrs. Vishakha Rane** HOD name**: Dr.Md.Imaduddin**
Signature: Signature:

Signature: ----------
Name : Dr. Kamal Shah
Principal,
St. John College of Engineering and Management.

Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Smit Patel [EU2237021]----------

Swikar Patil [EU2237006]----------

Aaditya Singh [EU1237012]----------

Ayush Varma [EU2237015] ----------

Date:

Acknowledgement

We express our sincere gratitude to all those who have directly and indirectly contributed to the successful preparation of this project synopsis.

We are deeply grateful to **Dr. Kamal Shah, Principal, St. John College of Engineering and Management**, for providing an excellent academic environment and the necessary facilities to undertake this project.

We extend our heartfelt thanks to the **Dr. Md. Imaduddin, Head of the Department of Electronics and computer science Engineering** for their constant encouragement, valuable suggestions, and continuous support throughout the preparation of this synopsis.

We would like to express our sincere and heartfelt appreciation to our **project guide**, **Prof. Vishakha Rane**, for her invaluable guidance, constructive suggestions, technical expertise, and continuous motivation. Her encouragement and insightful feedback have been instrumental in shaping this project synopsis.

We are especially thankful to our **industry mentor**, **Mr. Rahul Gupta, IOFT**, for his valuable guidance, practical insights, and technical support. His expertise and timely suggestions have greatly helped us in understanding real-world requirements and shaping this project.

We also **acknowledge all the faculty members and laboratory staff** of **the Department of Electronics and Computer Science Engineering** for their timely assistance, cooperation, and technical support.

Finally, we express our heartfelt gratitude to our parents, family members, friends, and classmates for their constant encouragement, moral support, and cooperation throughout this work.

Group Members

1. Smit Patel [EU2237021]
2. Swikar Patil [EU2237006]
3. Aaditya Singh [EU1237012]
4. Ayush Varma [EU2237015]

ABSTRACT

Humanoid and service robots are increasingly being integrated with Artificial Intelligence to enable natural Human-Robot Interaction. Conventional robots generally depend on predefined commands, programmed sequences or dedicated control interfaces. Such systems provide reliable control but have limited flexibility when users want to interact with robots using natural language.

This project presents the **IOFT Humanoid Robot Framework**, an integrated electronics, embedded systems and Artificial Intelligence platform that combines local Large Language Model reasoning, Retrieval-Augmented Generation and a fail-closed hardware safety mechanism.

The computational system uses **Qwen 2.5 3B**, deployed locally using Ollama. A Retrieval-Augmented Generation system using **ChromaDB** provides project-specific knowledge, while **MongoDB** maintains multi-turn conversation history. A **FastAPI** backend manages communication between the user interface, AI system and embedded robot controller.
The AI-generated command does not directly control the motors. Instead, it passes through a safety validation layer that checks the tool, parameters, authority mode and telemetry freshness before allowing hardware execution.

The validated command is transmitted through **UDP communication to an ESP32**, which controls the motors and actuators through appropriate driver circuits. Sensors provide feedback and telemetry to the AI server. A hardware watchdog provides an additional safety mechanism by disabling motor operation when communication is interrupted.
Thus, the project combines **embedded electronics, motor control, sensors, wireless communication, local AI, RAG, database systems and software safety** to develop a safer and more intuitive intelligent robotic platform.

CONTENTS

- List of Figures i
- List of Tables ii
- Abbreviations and symbols iii

Chapter No.	Topic	Page No.
Chapter 1	Introduction: Background, Problem Statement, Need, Objectives, Scope, Expected Outcomes	**10**
Chapter 2	Introduction, Literature Review Table, Research Gap, Novelty/Innovation	**14**
Chapter 3	Proposed Methodology: Working Principle, Architecture/Block Diagram, DFD Flowchart, Hardware And Software Methodology, Components	**17**
Chapter 4	Project Planning: Wbs, Gantt Chart, Roles, Risk Assessment	**27**
Chapter 5	Resources & Feasibility: Technical/Economic/Operational Feasibility, Budget, Bom	**30**
Chapter 6	Expected Results	**32**
Chapter 7	Sustainability And Impact: Industry Relevance, Societal Impact, Environmental Impact, Sustainable Development Goals	**34**
Chapter 8	Conclusion, References And Appendix	**36**

List of Figures

Figure No.	Figure Name	Page No.
Figure 1	Overall System Architecture	18
Figure 2	Hardware Block Diagram	20
Figure 3	AI and RAG Architecture	21
Figure 4	Level-0 Data Flow Diagram	21
Figure 5	Command Safety Flowchart	22
Figure 6	ESP32 Motor Control Architecture	23
Figure 7	Complete AI-to-Motor Pipeline	39

List of tables

Table No.	Title of Table	Page No.
Table 1	Literature Review And Comparative Analysis	14-15
Table 2	Hardware Components And Specifications	26
Table 3	Software Components And Technologies	26
Table 4	Gantt Chart	28
Table 5	Roles And Responsibilities	29
Table 6	Risk Assessment	29
Table 7	Budget Estimation	31
Table 8	Bill Of Materials	31
Table 9	Expected System Performance	33
Table 10	SDG Mapping	35
Table 11	Communication Parameters	38

Abbreviations and symbols

- **ML**: Machine Learning
- **AI**: Artificial Intelligence
- **TF-IDF**: Term Frequency-Inverse Document Frequency
- **NLP**: Natural Language Processing
- **DFD**: Data Flow Diagram
- **UML**: Unified Modeling Language
- **CSV**: Comma-Separated Values
- **UI**: User Interface
- **PDF**: Portable Document Format
- **HR**: Human Resources
- **ETL**: Extract, Transform, Load

CHAPTER 1: INTRODUCTION

1.1 Background

The integration of Artificial Intelligence with robotics has created new possibilities for Human-Robot Interaction. Conventional robots generally operate using predefined programs, remote controllers or fixed sequences of commands. Although these methods provide predictable operation, they require the user to understand the robot's control interface.

Large Language Models provide an alternative approach by allowing users to communicate with robotic systems using natural language.

For example, instead of directly controlling a motor through a controller, an operator can provide a high-level instruction. The AI system interprets the instruction, converts it into a structured command and passes the command to the embedded controller.

However, direct connection between an LLM and physical hardware introduces safety problems. An LLM may produce invalid commands, unexpected parameters or commands that do not match the actual hardware configuration.

The proposed project addresses this problem by placing an independent safety validation layer between the AI system and the ESP32 hardware controller.

1.2 PROBLEM STATEMENT

Existing natural-language robotic systems have the following problems:

1. Unconstrained AI Output

An AI model may generate commands or parameters that are not suitable for the physical robot.

2. Lack of Hardware Awareness

A general-purpose language model does not automatically understand the exact motors, sensors, voltage requirements and operating limitations of a particular robot.

3. Communication Failure

Loss of communication between the AI server and embedded controller can cause unsafe or incomplete operation.

4. Lack of Independent Safety Validation

If the LLM itself is responsible for determining whether its own command is safe, an incorrect AI output may reach the hardware.

5. Limited Natural Interaction

Traditional robots generally require specific commands, joysticks or dedicated control interfaces.

Therefore, the proposed project aims to provide natural-language interaction while keeping

the AI system separated from direct motor control through an independent validation layer.

1.3 NEED OF THE PROJECT

The project is required because modern robotic systems need both **intelligent interaction and reliable hardware control**.

The major requirements are:

- Natural-language interaction.
- Local AI processing.
- Low dependence on cloud services.
- Project-specific knowledge retrieval.
- Conversation memory.
- ESP32-based control.
- Motor and actuator control.
- Sensor-based feedback.
- Wireless communication.
- Command validation.
- Watchdog protection.
- Hardware/software integration.

The original project document identifies the requirement for local AI, safety validation and microcontroller-level failsafe protection.

1.4 OBJECTIVES

General Objective

To design and develop an intelligent humanoid robot capable of understanding natural-language instructions using a locally deployed language model and safely converting those instructions into validated commands for embedded hardware.

Specific Objectives

1. To develop an ESP32-based embedded robotic control system.
2. To interface motors and actuators with the ESP32 using suitable driver circuits.
3. To integrate sensors for collecting robot-state information.
4. To implement UDP-based communication between the AI server and ESP32.
5. To deploy Qwen 2.5 3B locally using Ollama.
6. To adapt the LLM for structured robotic tool calling.
7. To implement Retrieval-Augmented Generation using ChromaDB.
8. To implement conversation memory using MongoDB.
9. To develop a FastAPI backend.
10. To develop an independent fail-closed safety validation layer.

11. To validate AI-generated commands before hardware execution.
12. To implement watchdog-based hardware protection.
13. To evaluate the complete AI-to-hardware communication pipeline.

These objectives retain the AI, embedded, RAG and safety objectives from the supplied document while removing the mechanical/CAD-specific objectives. The original document specifies ESP32 firmware, Qwen 2.5 3B, RAG, safety validation and experimental validation as the core objectives.

1.5 SCOPE

Hardware Scope

The hardware portion includes:

- ESP32 controller.
- Motors.
- Motor driver.
- Servo/actuator control.
- IMU sensor.
- Ultrasonic sensors.
- Power supply.
- DC-DC buck converters.
- Robot electronic wiring.
- Communication interface.

Software Scope

The software portion includes:

- Qwen 2.5 3B.
- Ollama.
- FastAPI.
- MongoDB.
- ChromaDB.
- Sentence-transformer embeddings.
- RAG.
- Command validation.
- UDP communication.
- Telemetry monitoring.
- Web/VR interface.

The original architecture specifies ESP32 command communication on **Port 8888** and telemetry on **Port 8889**.

Limitations

- The robot is primarily intended for controlled environments.
- Local LLM performance depends on available computing resources.
- The current system does not implement advanced visual SLAM.
- The robot's operation depends on the available motor, actuator and sensor hardware.
- The current safety system validates software commands but does not replace physical safety supervision.

Future Scope

Future improvements can include:

- Depth cameras.
- Computer vision.
- Visual SLAM.
- Better obstacle detection.
- More advanced motor control.
- Additional sensors.
- Improved AI planning.
- More advanced robotic manipulation.

1.6 EXPECTED OUTCOMES

The project is expected to produce:

1. A functional intelligent humanoid robot prototype.
2. Natural-language interaction.
3. Local Qwen 2.5 3B inference.
4. RAG-based project knowledge retrieval.
5. MongoDB-based conversation memory.
6. Structured AI tool calling.
7. Independent command validation.
8. ESP32-based motor and actuator control.
9. Sensor telemetry.
10. Watchdog-based protection.
11. Safe AI-to-hardware communication.

CHAPTER 2: LITERATURE REVIEW

2.1 INTRODUCTION

The convergence of Artificial Intelligence and robotics has resulted in language-driven robotic systems capable of interpreting high-level human instructions.

Research has explored language-based planning, embodied AI, RAG, efficient LLM fine-tuning and safe robotic control.

However, there is still a requirement for an integrated architecture that combines:

Natural Language → Local LLM → RAG → Memory → Safety Validation → ESP32 → Motor Control → Sensor Feedback

2.2 LITERATURE SURVEY

Table 1: Literature Survey

Paper / Reference	Year	Methodology	Key Finding	Research Gap
SayCan: Grounding Language in Robotic Affordances	2022	LLM reasoning combined with robotic affordances	Enables language-based robotic task planning	Does not provide the proposed independent hardware safety gateway
Code as Policies	2023	LLM generates robotic control programs	Enables flexible robotic control	Generated code can introduce safety risks
Inner Monologue	2022	Uses environment feedback in language-based planning	Improves recovery and task reasoning	Does not provide the complete proposed fail-closed hardware layer
Retrieval-Augmented Generation	2020	Retrieves relevant knowledge before generation	Improves knowledge-grounded responses	Original work focuses on NLP rather than embedded robot control
QLoRA	2023	Quantized parameter-efficient fine-tuning	Reduces memory requirements	Not specifically designed for robotic command validation

Control Barrier Functions	2019	Mathematical safety constraints	Provides formal safety properties	Complex to combine directly with conversational AI
ROS-LLM	2023	Language model connected with robotic middleware	Enables language-driven robot actions	Cloud dependency and latency concerns
Bipedal Locomotion using RL	2021	Reinforcement learning for locomotion	Improves robotic walking	Does not focus on natural-language interaction
Safe Reinforcement Learning	2015	Constrained learning	Reduces unsafe actions	Does not address LLM command validation
Human-Robot Interaction through AR/VR	2019	Mixed-reality robotic interaction	Improves operator awareness	Does not combine conversational memory and local LLM

2.3 RESEARCH GAP

Gap 1: Lack of Independent Safety Validation

Many language-driven robots depend on the LLM following instructions correctly. The proposed system introduces an independent safety validator.

Gap 2: Cloud Dependency

Cloud-based models require network connectivity and may introduce latency.

Gap 3: Lack of Project-Specific Knowledge

A general LLM does not automatically know the project's hardware and software configuration. RAG addresses this problem.

Gap 4: Lack of Persistent Conversation

MongoDB allows conversation history to be maintained across interactions.

Gap 5: Lack of Hardware Communication Protection

The ESP32 watchdog provides an additional protection mechanism if communication is interrupted.

2.4 NOVELTY / INNOVATION

The proposed project provides the following innovations:

1. Local AI

Qwen 2.5 3B runs locally instead of depending entirely on cloud APIs.

2. RAG-Based Knowledge

ChromaDB provides project-specific knowledge retrieval.

3. Persistent Memory

MongoDB maintains conversation history.

4. Fail-Closed Safety

Invalid commands are rejected before reaching the robot.

5. Telemetry-Based Validation

Hardware execution depends on valid and sufficiently recent telemetry.

6. ESP32 Watchdog

The embedded controller provides an additional protection mechanism.

7. AI + Embedded Integration

The project integrates:

AI + Software + Electronics + Motors + Sensors + Communication

CHAPTER 3: PROPOSED METHODOLOGY

3.1 PROPOSED SYSTEM

The proposed system consists of three major layers.

Layer 1 — Human Interaction

The user interacts through:

- Web interface
- Voice input
- VR interface

Layer 2 — AI and Software

The local computer contains:

- FastAPI
- Qwen 2.5 3B
- Ollama
- ChromaDB
- MongoDB
- RAG system
- Safety validator

Layer 3 — Embedded Electronics

The ESP32 controls:

- Motors
- Actuators
- Sensors
- PWM signals
- Telemetry
- Watchdog protection

FIGURE 1: OVERALL SYSTEM ARCHITECTURE

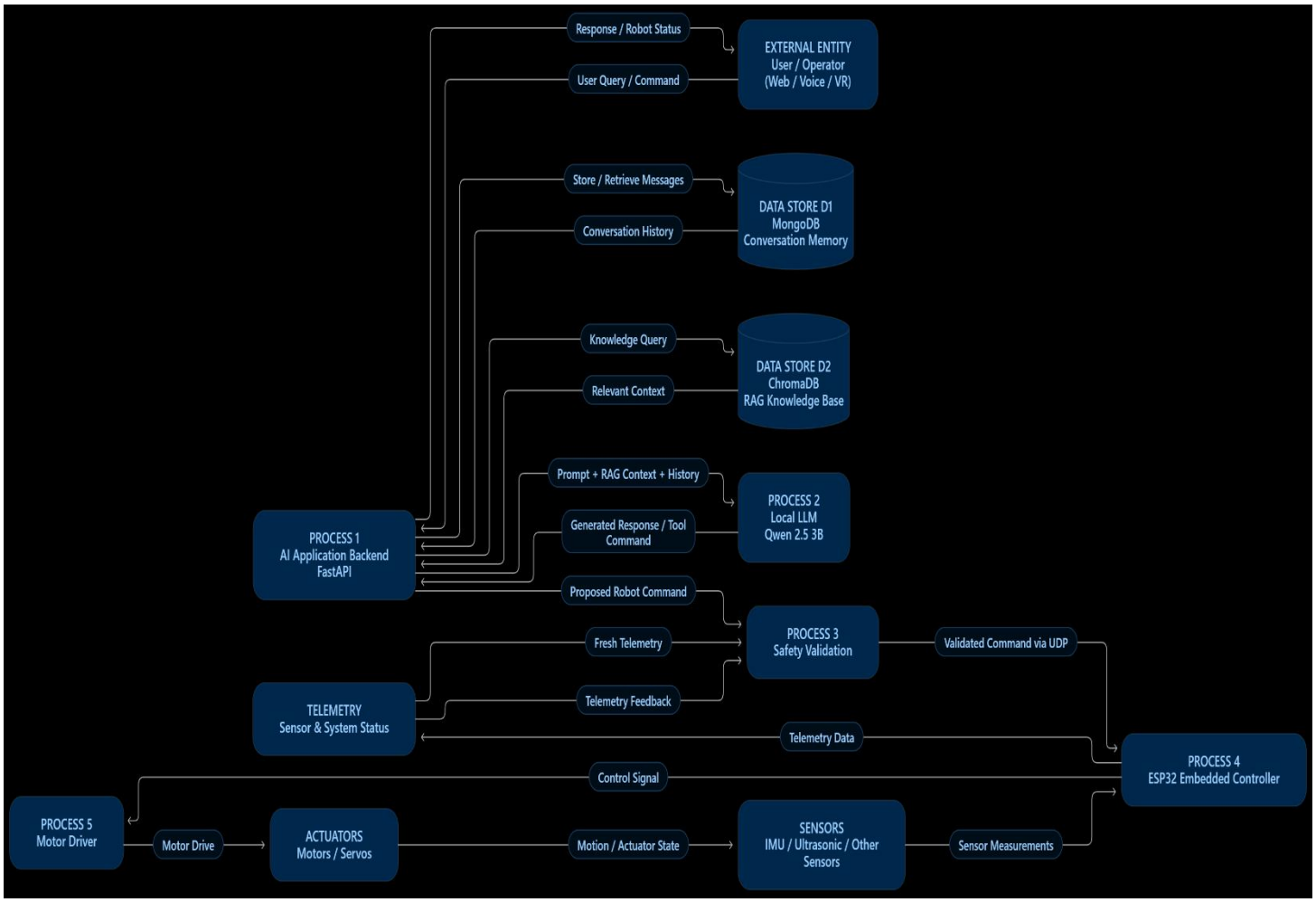


Figure 1: Overall System Architecture

3.2 WORKING PRINCIPLE

The system works in the following sequence:

Step 1: User Input

The operator enters a natural-language instruction.

Step 2: Conversation Retrieval

Previous relevant conversation information is retrieved from MongoDB.

Step 3: Knowledge Retrieval

If project-specific information is required, relevant information is retrieved from ChromaDB.

Step 4: AI Processing
Qwen 2.5 3B processes:
System Prompt + Conversation History + RAG Context + User Input

Step 5: Structured Command Generation
The LLM generates a structured command.

Step 6: Safety Validation
The command is checked for:

- Valid tool
- Correct parameters
- Allowed parameter range
- Correct authority
- Valid telemetry

Step 7: UDP Communication
Only after successful validation is the command sent to ESP32.

Step 8: Motor Control
ESP32 processes the command and generates appropriate control signals for the motor driver.

Step 9: Sensor Feedback
Sensors continuously provide robot-state information.

Step 10: Telemetry
Telemetry is transmitted back to the AI server.
The supplied document specifies the same six-stage working principle and UDP packet architecture.

3.3 HARDWARE BLOCK DIAGRAM

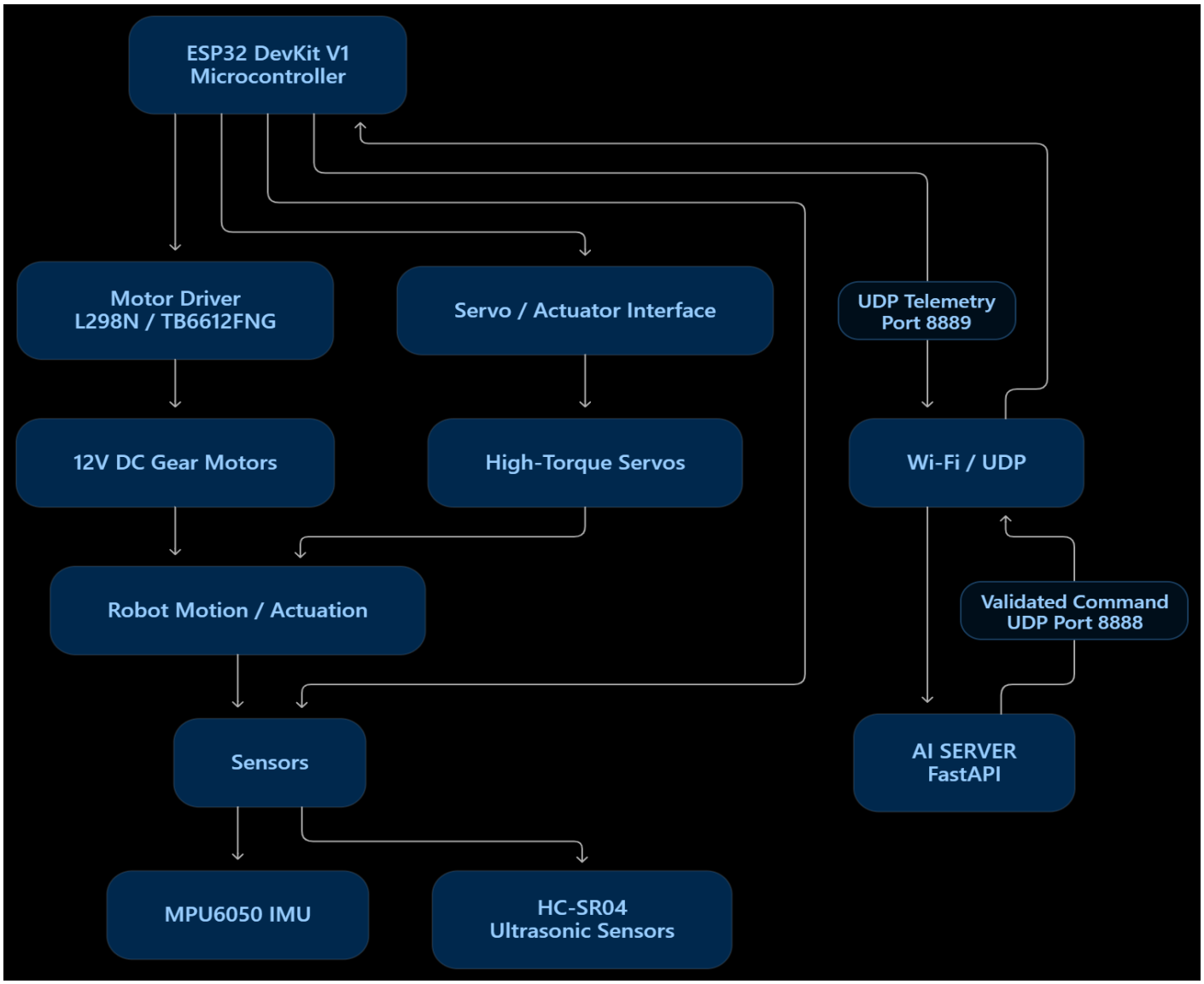


Figure 2: Hardware Block Diagram

3.4 AI + RAG ARCHITECTURE

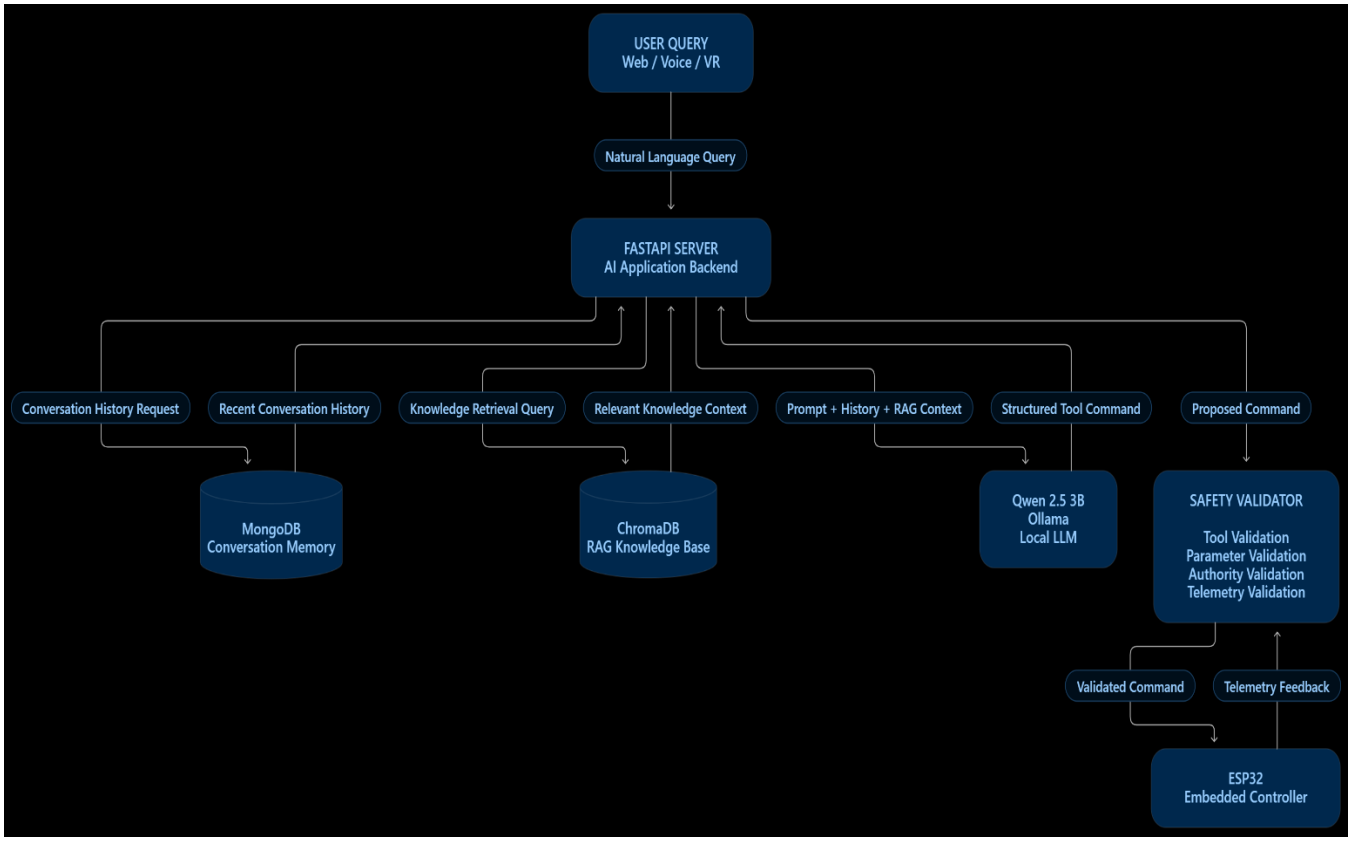


Figure 3: AI, RAG and Conversation Memory Architecture

3.5 DATA FLOW DIAGRAM

Level 0 DFD

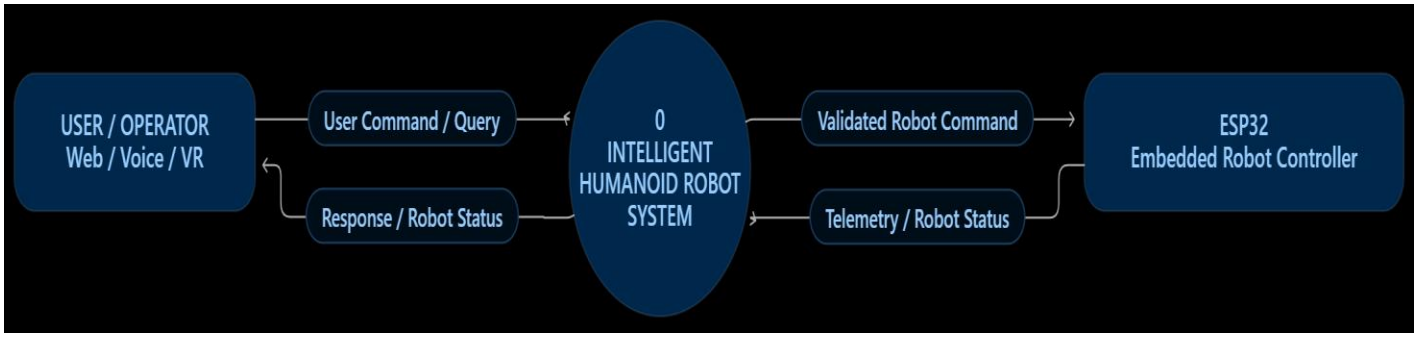


Figure 4: Level-0 Data Flow Diagram

3.6 COMMAND SAFETY FLOWCHART

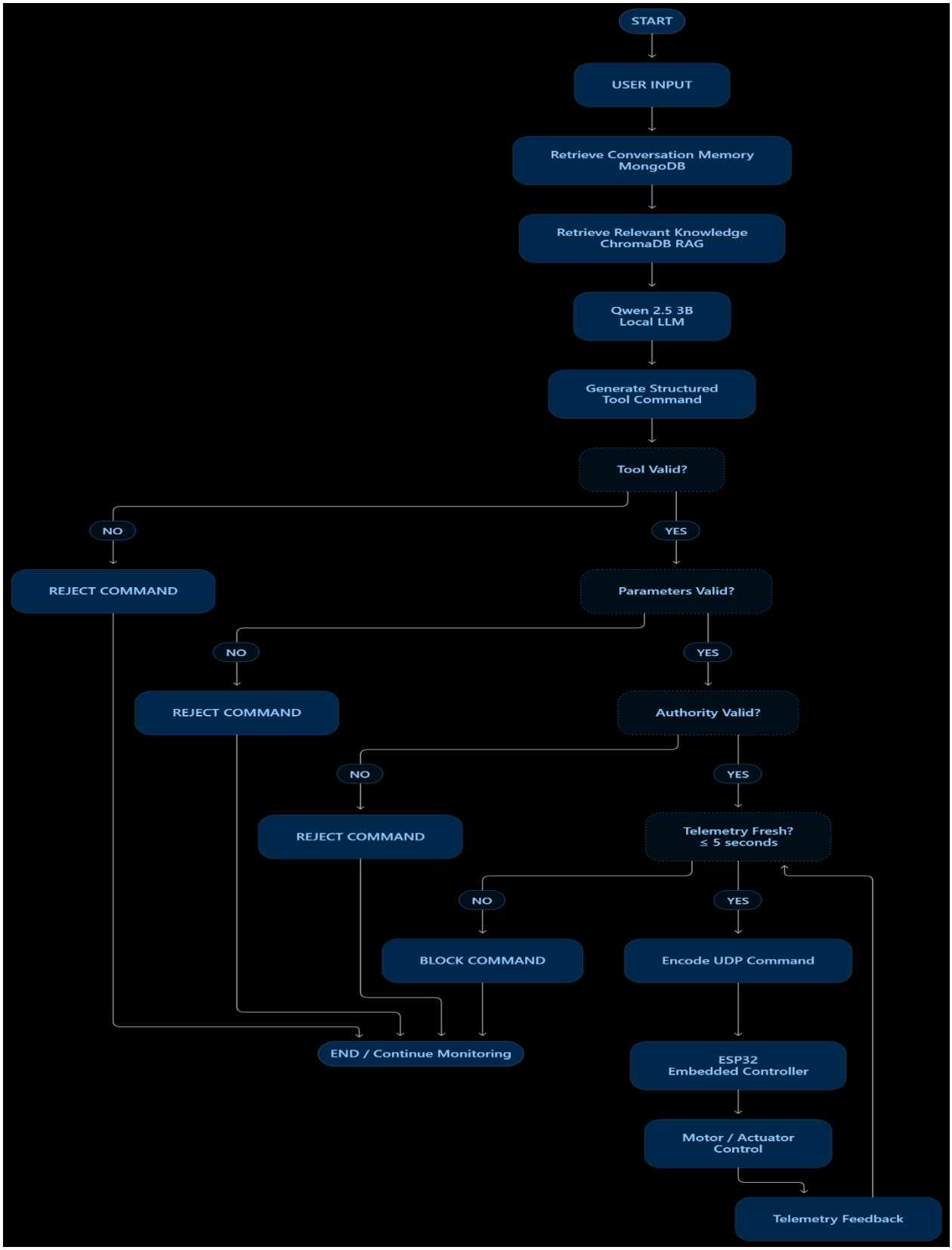


Figure 5: Fail-Closed Safety Validation Flowchart

3.7 MOTOR CONTROL METHODOLOGY

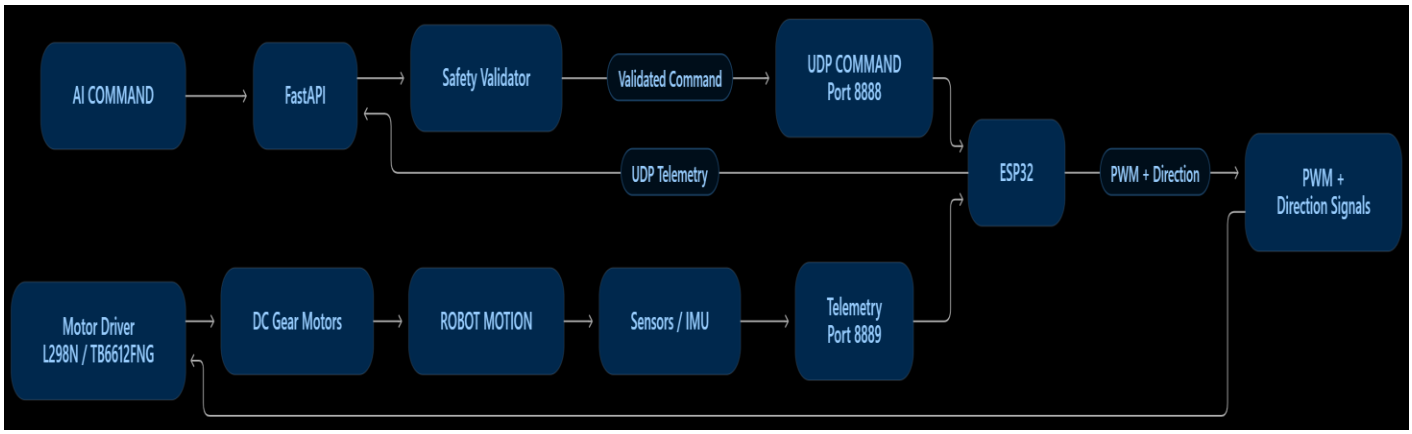


Figure 6: ESP32 Motor Control Architecture

Motor Control Process

1. The user provides a natural-language command.
2. Qwen 2.5 3B interprets the command.
3. The generated command is validated.
4. The validated command is converted into a predefined communication format.
5. The command is transmitted to the ESP32 through UDP.
6. ESP32 receives the command.
7. ESP32 generates PWM and direction control signals.
8. Motor driver provides the required current to the motor.
9. Motor produces the required movement.
10. Sensors provide feedback to the ESP32.
11. ESP32 sends telemetry to the AI server.

The supplied protocol uses the format:
<COMMAND>|<SPEED_PWM>|<DURATION_SEC>
For example:
FORWARD|20|1.00
and the expected acknowledgement is:
MOTOR_ACK

3.8 SENSOR METHODOLOGY

The project uses sensors to provide information about the physical state of the robot.

MPU6050

The MPU6050 provides:

- Accelerometer data
- Gyroscope data
- Motion information
- Orientation-related telemetry

HC-SR04

The ultrasonic sensor provides:

- Distance measurement
- Obstacle detection
- Forward collision awareness

The supplied component specification lists one MPU6050 and two HC-SR04 sensors.

3.9 SOFTWARE METHODOLOGY

AI Layer

Qwen 2.5 3B

Used for:

- Natural language understanding
- Intent interpretation
- Structured tool calling
- Robot-related reasoning

Model Deployment

Ollama

Used to run the model locally.

RAG Layer

ChromaDB + Sentence Transformers

Used for:

- Document storage
- Embeddings
- Similarity search
- Project-specific knowledge retrieval

Memory Layer

MongoDB

Used for:

- Conversation sessions
- Previous messages
- Context retrieval

Backend

FastAPI

Used for:

- Chat API
- Command API
- Robot state
- Assistant interface
- Communication with the robot

Safety Layer

Used for:

- Tool allowlisting
- Parameter validation
- Authority verification
- Telemetry freshness
- Command rejection

3.10 QLoRA METHODOLOGY

The supplied project specifies QLoRA-based adaptation of Qwen 2.5 3B.

The approach uses quantized model weights and trainable low-rank adapters.

The supplied document specifies:

- Rank: **r = 16**
- Alpha: **α = 32**
- Target layers:
  - q_proj
  - k_proj
  - v_proj
  - o_proj
- Training data: **over 1,500 domain-specific tool-calling examples.**

3.11 COMPONENTS AND MATERIALS

Table 2: Hardware Components

Component	Specification	Quantity	Function
ESP32 DevKit V1	Dual-core 240 MHz, Wi-Fi	1	Main embedded controller
High-Torque Metal Gear Servo	MG996R / RDS3115	6	Actuator control
DC Planetary Gear Motor	12V High-Torque	2	Robot locomotion
Motor Driver	IBT-2	1	Motor direction and PWM control
Battery	11.1V 3S 2200mAh	1	Power source
LM2596 Buck Converter	Adjustable 3A	2	Voltage regulation
Robot Chassis	Project-specific structure	1 Set	Supports electronics and actuators

TABLE 3: SOFTWARE COMPONENTS

Software	Purpose
Python	AI and backend development
FastAPI	Backend API
Ollama	Local LLM deployment
Qwen 2.5 3B	Language reasoning
ChromaDB	Vector database
Sentence Transformers	Text embeddings
MongoDB	Conversation memory
Pydantic	Data and command validation
UDP	Robot communication
ESP32 Firmware	Embedded control

CHAPTER 4: PROJECT PLANNING

4.1 Work Breakdown Structure (WBS)

- Phase 1: Research & Kinematic System Design
  - 1.1 Requirements specification and safety protocol definition
  - 1.2 SolidWorks CAD design of humanoid frame, manipulator linkages, and electronics bays
  - 1.3 Kinematic modeling and joint angular velocity limit derivation
- Phase 2: Embedded Mechatronics & Firmware Development
  - 2.1 ESP32 circuit layout, buck regulation, and hardware breadboarding
  - 2.2 FreeRTOS firmware programming: dual-port UDP listener and 1000 ms watchdog
  - 2.3 Sensor integration (MPU6050 IMU, HC-SR04 ultrasonic) and telemetry packet packing
- Phase 3: AI Engine, RAG & Safety Software Architecture
  - 3.1 FastAPI server implementation, Pydantic schemas, and MongoDB session manager
  - 3.2 ChromaDB vector store configuration with document chunking and embeddings
  - 3.3 Dataset curation and QLoRA fine-tuning of Qwen 2.5 3B for deterministic tool calling
  - 3.4 Authoritative fail-closed validator (CommandSafetyValidator) implementation
- Phase 4: Hardware Assembly, System Integration & Testing
  - 4.1 3D printing of chassis components and physical mechanical assembly
  - 4.2 Network loopback and latency profiling between FastAPI server and ESP32
  - 4.3 End-to-end testing across 266 validation prompts covering boundary violations and recovery

4.2 GANTT CHART

Table 4: Project Planning Schedule (M= month)

Activity	M 1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12	M13
Literature Survey	✓												
Problem Definition	✓												
Hardware Selection	✓	✓											
ESP32 Firmware		✓	✓										
Motor Integration		✓	✓										
Sensor Integration		✓	✓										
LLM Dataset Preparation			✓	✓									
QLoRA Fine-Tuning			✓	✓									
RAG Development			✓	✓									
MongoDB Memory				✓									
FastAPI Backend				✓									
Safety Validator				✓	✓								
Hardware-Software Integration					✓								
Testing					✓	✓							
Documentation						✓							
Research Paper							✓	✓	✓	✓	✓	✓	✓

4.3 ROLES AND RESPONSIBILITIES

Table 5: Team Roles

Student	Role	Responsibilities
Smit Patel	Embedded Systems	ESP32 firmware, motor driver interface, sensors and communication
Swikar Patil	Hardware Integration	Motor/actuator wiring, power system, sensor integration and hardware testing
Aaditya Singh	AI/LLM	Qwen 2.5 3B, Ollama, QLoRA and structured tool calling
Ayush Varma	Software & Safety	FastAPI, RAG, MongoDB, validation and testing

4.4 RISK ASSESSMENT

Table 6: Risk Assessment and Mitigation

Risk	Severity	Likelihood	Mitigation
Invalid AI command	High	Medium	Command validation
Incorrect parameter	High	Medium	Pydantic validation
Motor overload	High	Medium	Parameter and hardware limits
Wireless packet loss	High	Medium	ESP32 watchdog
Telemetry loss	High	Medium	Freshness validation
Power fluctuation	Medium	Medium	Regulated power supply
LLM memory limitation	Medium	Low	Quantized model
Sensor failure	Medium	Medium	Telemetry monitoring
Unexpected AI output	High	Low	Fail-closed rejection

CHAPTER 5: RESOURCES AND FEASIBILITY

5.1 TECHNICAL FEASIBILITY

The project is technically feasible because it uses accessible embedded hardware and open-source software.

The ESP32 provides:

- Dual-core processing
- Wi-Fi communication
- PWM capability
- Embedded control

The software uses:

- Python
- FastAPI
- Qwen 2.5 3B
- Ollama
- ChromaDB
- MongoDB
- Pydantic

The supplied project estimates approximately **3.2 GB VRAM** for Qwen 2.5 3B under 4-bit quantization.

5.2 ECONOMIC FEASIBILITY

The system uses open-source software and locally deployed AI.

This reduces dependency on recurring cloud API costs.

The supplied budget estimates the prototype at approximately:

₹12,950

5.3 OPERATIONAL FEASIBILITY

The system is designed so that the operator does not need to directly control individual motor signals.

Instead:

Natural Language ->AI Interpretation ->Command Validation ->ESP32 ->Motor Control

This makes the robot easier to operate and demonstrates the integration of AI with embedded electronics.

5.4 BUDGET ESTIMATION

Table 7: Budget

No.	Component	Quantity	Cost
1	ESP32	1	₹550
2	MG996R Servo	6	₹2,400
3	DC Planetary Gear Motor	2	₹1,600
4	Motor Driver	1	₹280
5	MPU6050	1	₹220
6	HC-SR04	2	₹240
7	LiPo Battery	1	₹1,850
8	LM2596	2	₹260
9	Printing Material	2 Spools	₹2,100
10	Fasteners/Bearings	1 Kit	₹750
11	PCB/Wires/Terminals	1 Kit	₹450
12	Battery Charger	1	₹750
13	Miscellaneous	—	₹1,500
	TOTAL		**₹12,950**

5.5 BILL OF MATERIALS

Table 8: BOM

Component	Specification	Quantity	Function
ESP32	Dual Core 240 MHz	1	Controller
MG996R Servo	High torque	6	Actuation
DC Gear Motor	12V high torque	2	Locomotion
Motor Driver	IBT-2	1	Motor control
MPU6050	6-axis IMU	1	Motion sensing
HC-SR04	Ultrasonic	2	Distance sensing
LiPo Battery	11.1V 2200mAh	1	Power
LM2596	3A adjustable	2	Voltage regulation
Wires/PCB	Prototyping components	1 Kit	Electrical connections

CHAPTER 6: EXPECTED RESULTS

Expected Result 1 — Command Validation
The system is expected to reject malformed or invalid commands before they reach the ESP32.

Expected Result 2 — Safe Hardware Communication
Only validated commands should be transmitted to the robot.

Expected Result 3 — Local AI Response
The system is expected to provide local AI inference without requiring continuous cloud API access.

Expected Result 4 — RAG Retrieval
The system is expected to retrieve project-specific information from the ChromaDB knowledge base.

Expected Result 5 — Conversation Memory
MongoDB is expected to maintain relevant conversation history.

Expected Result 6 — Telemetry Monitoring
The system is expected to receive robot-state information from the ESP32.

Expected Result 7 — Watchdog Protection
The system is expected to stop motor operation when the communication watchdog condition is triggered.

Expected Result 8 — AI Intent Recognition
The system is expected to correctly identify the user's intended robot operation and convert it into a structured command.

TABLE 9: EXPECTED SYSTEM PERFORMANCE

Parameter	Expected Target
Schema Compliance	100%
Unexpected UDP Dispatches	0 during negative tests
AI Response Latency	< 1500 ms target
Telemetry Freshness	≤ 5 seconds
Watchdog Timeout	1000 ms
Command Validation	100% invalid-command rejection
RAG Retrieval	Relevant project context
Conversation Memory	Persistent session-based

CHAPTER 7: SUSTAINABILITY AND IMPACT

7.1 INDUSTRIAL RELEVANCE

The proposed architecture can be extended to:

- Industrial automation
- Automated Guided Vehicles
- Collaborative robots
- Service robots
- Smart laboratory systems
- Educational robotics
- Assistive robotics

7.2 SOCIETAL IMPACT

Natural-language interaction can make robotic systems easier for non-technical users to operate.

Potential applications include:

- Educational robotics
- Assistive systems
- Laboratory automation
- Service robotics
- Human-Robot Interaction

7.3 ENVIRONMENTAL IMPACT

The use of a relatively small locally deployed language model reduces dependence on large cloud-based AI systems.

The project also uses a modular architecture where individual electronic components can be replaced or upgraded without redesigning the complete system.

7.4 Table 10: SDG MAPPING

SDG	Contribution
SDG 3 – Good Health and Well-Being	Potential assistive robotic applications
SDG 7 – Affordable and Clean Energy	Local and optimized computing
SDG 9 – Industry, Innovation and Infrastructure	AI and robotics innovation
SDG 11 – Sustainable Cities and Communities	Intelligent robotic assistance
SDG 12 – Responsible Consumption and Production	Modular and replaceable hardware
SDG 13 – Climate Action	Reduced dependency on large cloud infrastructure

CHAPTER 8: CONCLUSION

Problem Addressed

Traditional robotic systems require predefined commands and interfaces. At the same time, directly connecting generative AI models to physical hardware can create unsafe conditions because of incorrect or unpredictable AI outputs.

Proposed Solution

The proposed **IOFT Intelligent Humanoid Robot System** combines:

- Local Qwen 2.5 3B language reasoning.
- Ollama for local model deployment.
- ChromaDB-based RAG.
- MongoDB conversation memory.
- FastAPI backend.
- ESP32 embedded controller.
- Motor and actuator control.
- Sensor telemetry.
- UDP communication.
- Fail-closed safety validation.
- Hardware watchdog protection.

The architecture therefore creates a controlled path between natural-language AI and physical hardware:

Expected Contribution

The project demonstrates how **Artificial Intelligence, embedded electronics, communication systems and robotic hardware** can be integrated into a single platform while maintaining an independent safety layer between AI reasoning and physical actuation.

REFERENCES

1. M. Ahn, A. Brohan, N. Brown, Y. Chebotar, O. Cortes, B. David, et al., "Do as I can, not as I say: Grounding language in robotic affordances," *Proc. Conf. Robot Learn. (CoRL)*, 2022.
2. J. Liang, W. Huang, F. Xia, P. Xu, K. Hausman, B. Ichter, et al., "Code as Policies: Language Model Programs for Embodied Control," *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2023.
3. W. Huang, F. Xia, T. Xiao, H. Chan, J. Liang, P. Florence, et al., "Inner Monologue: Embodied Reasoning through Planning with Language Models," *Proc. Conf. Robot Learn. (CoRL)*, 2022.
4. P. Lewis, E. Perez, A. Piktus, V. Karpukhin, N. Goyal, et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," *Advances in Neural Information Processing Systems*, vol. 33, 2020.
5. T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "QLoRA: Efficient Finetuning of Quantized LLMs," *Advances in Neural Information Processing Systems*, vol. 36, 2023.
6. A. D. Ames, S. Coogan, M. Egerstedt, G. Notomista, K. Sreenath, and P. Tabuada, "Control Barrier Functions: Theory and Applications," *Proc. European Control Conference*, 2019.
7. F. Chen et al., "ROS-LLM / Embodied AI related framework," 2023.
8. J. Siekmann et al., "Sim-to-real transfer of blind bipedal locomotion for Cassie on rough terrain," *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2021.
9. J. Garcia and F. Fernandez, "A comprehensive survey on safe reinforcement learning," *Journal of Machine Learning Research*, vol. 16, 2015.
10. M. Walker, H. Hedayati, J. Lee, and D. Szafir, "Communicating robot motion intent with augmented reality," *Proc. ACM/IEEE Int. Conf. Hum.-Robot Interaction*, 2018.
11. B. Siciliano, L. Sciavicco, L. Villani, and G. Oriolo, *Robotics: Modelling, Planning and Control*. Springer.
12. J. J. Craig, *Introduction to Robotics: Mechanics and Control*. Pearson.
13. Espressif Systems, *ESP32 Technical Reference Manual*.
14. IEEE, *IEEE Standard for Robot Safety*, IEEE Std 1872-2015.
15. ISO, *ISO 13482: Robots and Robotic Devices — Safety Requirements for Personal Care Robots*, 2014

APPENDICES

A: ELECTRONICS AND SAFETY PARAMETERS

A.1 Communication Parameters

Parameter	Value
Command Protocol	UDP
ESP32 Command Port	8888
Telemetry Port	8889
Watchdog Period	1000 ms
Telemetry Freshness Timeout	5.0 s

APPENDIX B: UDP WIRE PROTOCOL

AI Server → ESP32

Format:

<COMMAND>|<SPEED_PWM>|<DURATION_SEC>

Example

FORWARD|20|1.00

Stop

STOP|0|0.00

Expected Acknowledgement

MOTOR_ACK

APPENDIX C: TELEMETRY FORMAT

The ESP32 sends telemetry to the AI server.

Example

```
{
  "roll": 0.12,
  "pitch": -0.04,
  "yaw": 182.4,
  "battery_v": 11.8,
  "us_distance_cm": 42.5
}
```

APPENDIX D: COMPLETE AI-TO-MOTOR PIPELINE

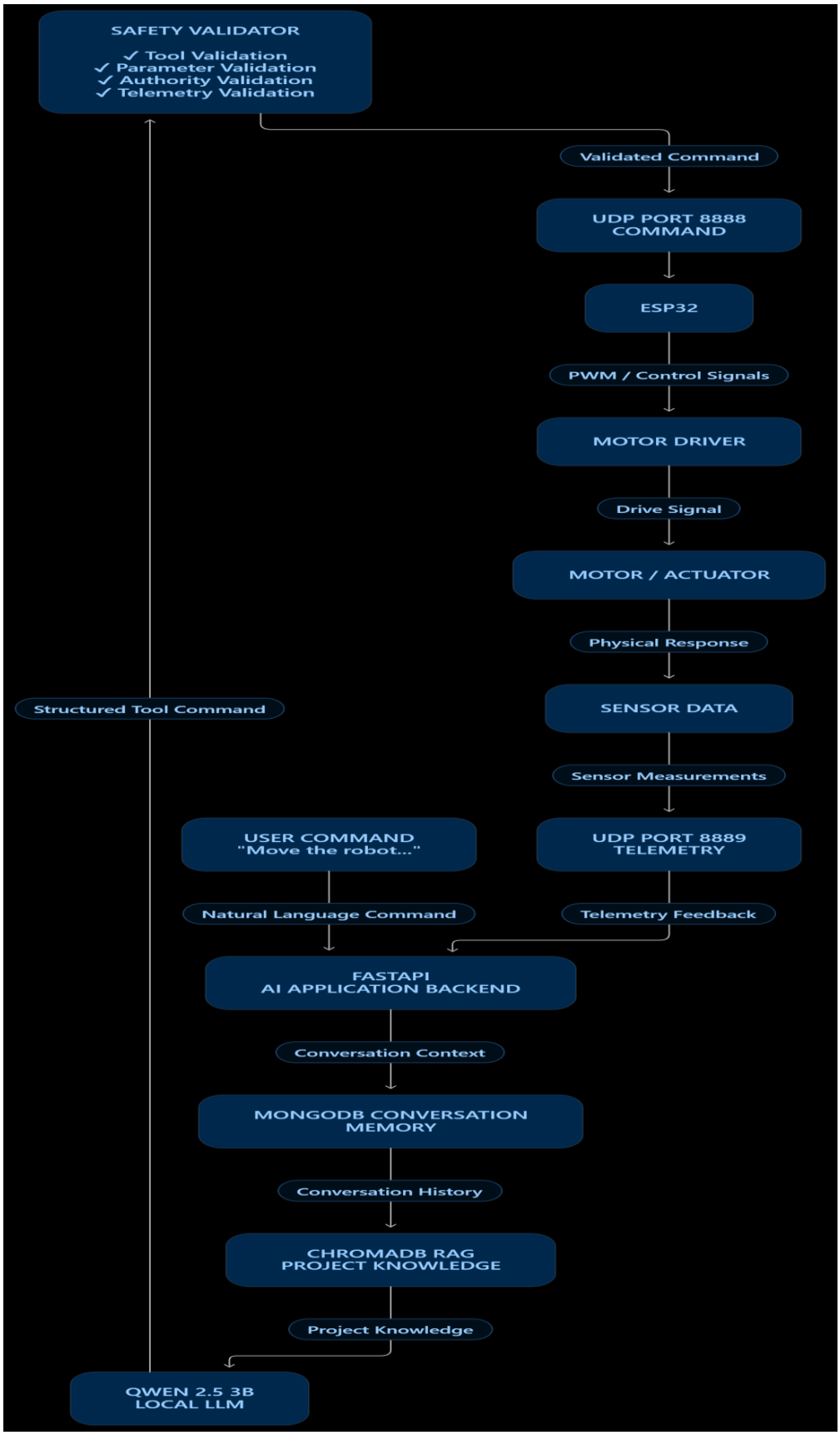


Figure: Complete AI-to-Hardware Closed-Loop Architecture